UNIVERSIDAD TECNICA
FEDERICO SANTA MARIA

Tesis de Magister

Design and Sizing of an Energy Storage System for a Hybrid Tugboat

Tesis para optar al título de
Magister en Ciencias de la Ingeniería Electrónico

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Enero 2025 , Valparaíso, Chile

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Agradecimientos

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Capítulo 1

Introduction

1.1. Motivation and Background

The growing global concern over climate change has significantly pressured industries to adopt sustainable practices. The maritime sector, responsible for approximately 2.5 % of energy sector CO_2 emmisions, has continuosly increased int emissions, as shown in Fig. 1.

https://theroundup.org/co2-greenhouse-gas-emission-statistics/

https://www.container-xchange.com/blog/shipping-emissions/

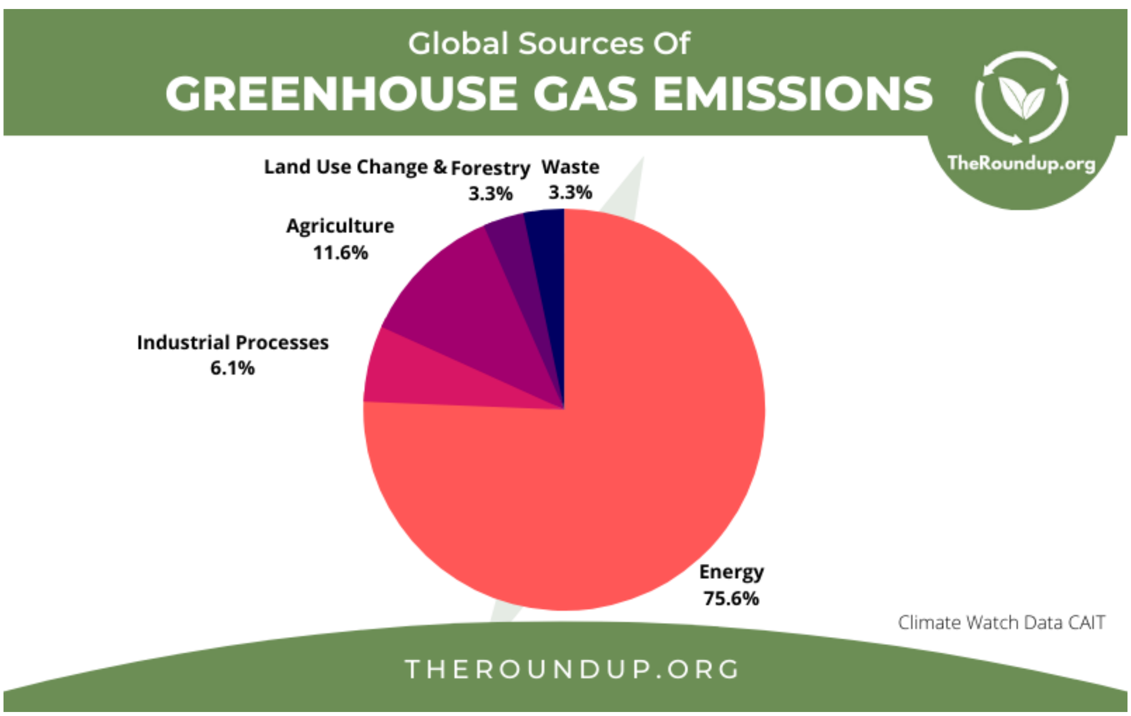
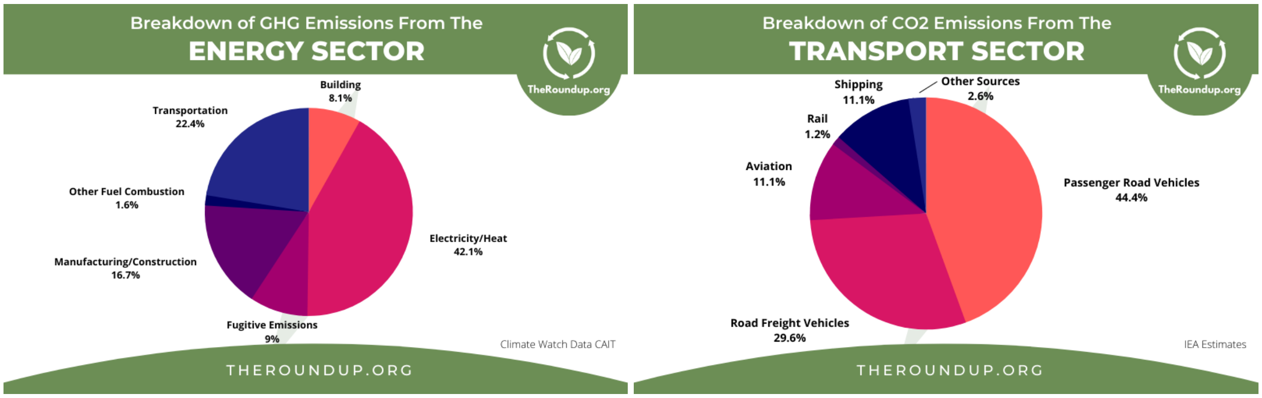


Figura 1.1: Global sources of greenhouse gas emissions. Source: theroundup.org

Addressing emission reduction is critical, as strict regulations on emissions and fuel efficiency aimed at mitigating the environmental impact of maritime activities are being implemented worldwide [2]. In the area of ships there are several activities to replace the proven diesel mechanical drives by new concepts. The triggers are firsty the ever invreasing price of fossil fuels and the other a mor strigent emission standard in so-called Sulphur Emission Control Areas



(a) CO_2 emissions from the energy sector. (b) CO_2 emissions from the transport subsector.

Figura 1.2: Breakdown of CO_2 emissions from the energy sector and the transport subsector. Source: theroundup.org

(SECA) from 1 January 2015. After that, only fuels may be used with a sulfur content of 0.1 percent in may coastal regions, which means additional costs for pushcase of fuel [2].

The International Maritime Organization (IMO) is implementing strategies to reduce the emissions from international shipping by establishing standards, imposing restrictions and provinding incentives. The initial strategy enisages a reduction in total GHG emissions from international shipping by at least 50 % by 2050 compared to 2008. consistent with the Paris Agreement. IMO's current global standard for SO_x emissions is 0.5 % mass by mass (m/m) by 2025 per the International Convention for the Prevention of Pollution by Ships (MARPOL 73/78) Annex VI. Similarly, the emission limit for NO_x from engines is dependent on the type of the engine and the nominal engine speed. The global limit has been set to 14.4 g/kWh for engines with speeds less than 130 rev/min and 7.7 g/kWh for engines with speed greater than 2000 rev/min.

To comply with the standards and emissions reduction goals set by IMO, the maritime industry is looking for reduced and zero-emission technologies and approaches. The transition from traditional fossil fuel engines to zero-emissions technologies is challenging in a competitive market, where manufacturers are struggling to increase revenue and reduce production cost. The incentives to reduce emissions and improve the efficiency of equipment are motivating ship manufacturers to adopt zero-emission technologies. Fuel cells and energy storages (for example, batteries and supercapacitors) reduce GHG emissions and are the key enablers for the realization of low-emissions operation of marine vessels.

Electric and hybrid propulsion architectures, allowed by innovative and hig performance components (batteries, fuel cells, supercapacitors, etc) are a promising track to reduce strongly pollutants and GHG emissions [3]. New developments of electrically propelled ships are based on an integrated energy distribution system architecture that releases large propulsion machines [4].

The main advantages of full-electric propulsion include better dynamic performance (start-up,arrest, changes in speed, and regenerative braking), reduced fuel consumption due to the more efficient and flexible use of generators that work within their minimum specific fuel consumption region, and the possibility of having shorter shaft lines because only the electric motor must be installed close to the propeller.

The hybrid propulsion system uses a combination of diesel engines or gas turbines with electric motors to provide a wide range of propeller speeds [4]. One of the main advantages of this configuration concerns an increase in efficiency at low loads. Hence, less fuel consumption,fewer

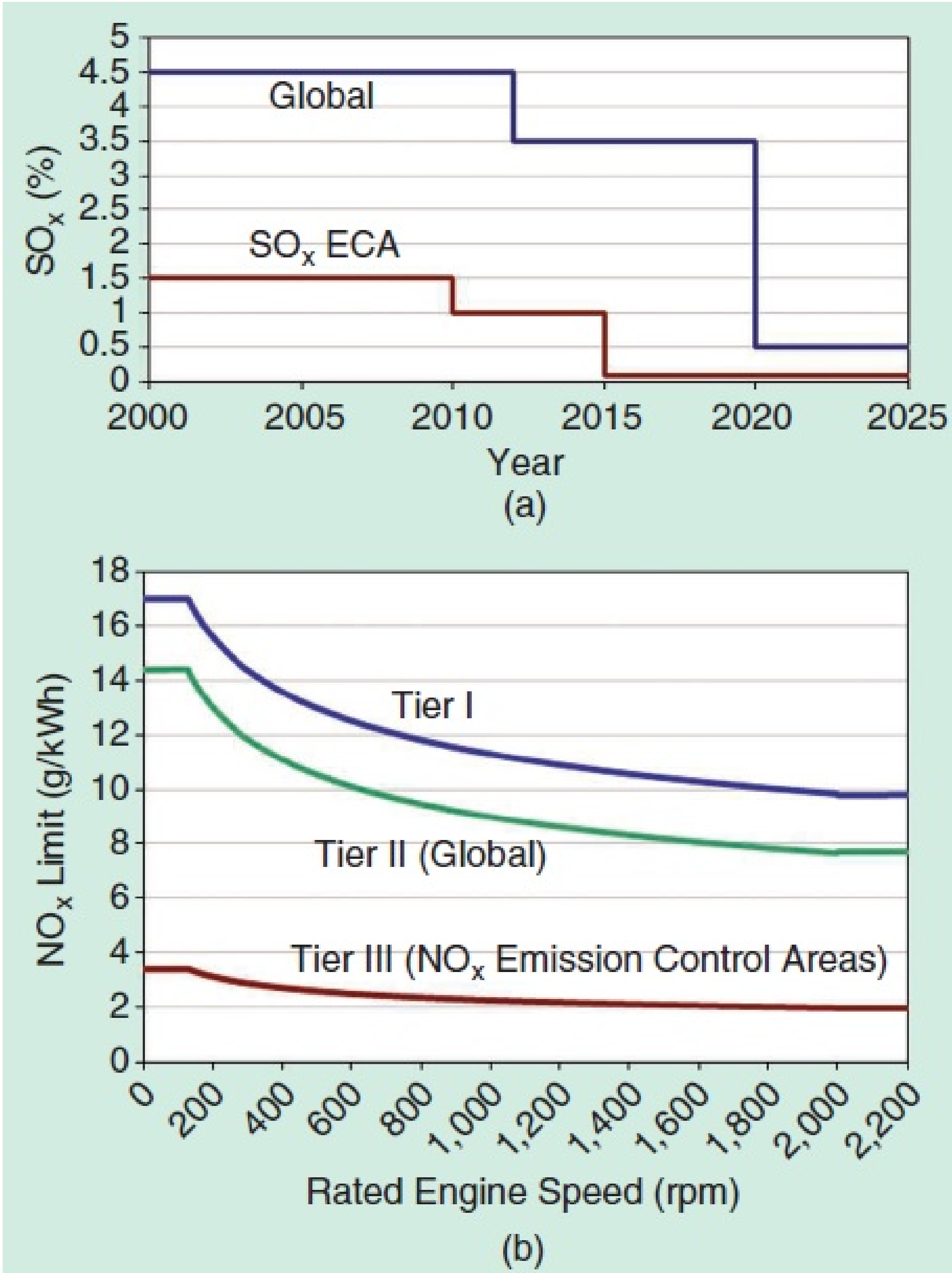


Figura 1.3: The MARPOL Annex VI emission limits for global and Emission Controlled Areas (ECAs perspective (a) SOx and (b) NOx. (Source: DieselNet

emissions, and a small footprint are achieved. Hybrid propulsion systems are interesting when there is a large variation in power demand during operations. The design rules of this type of systems can differ significatly for different types of vessel, mainly due to the different types of mission profiles and operation requirements [3].

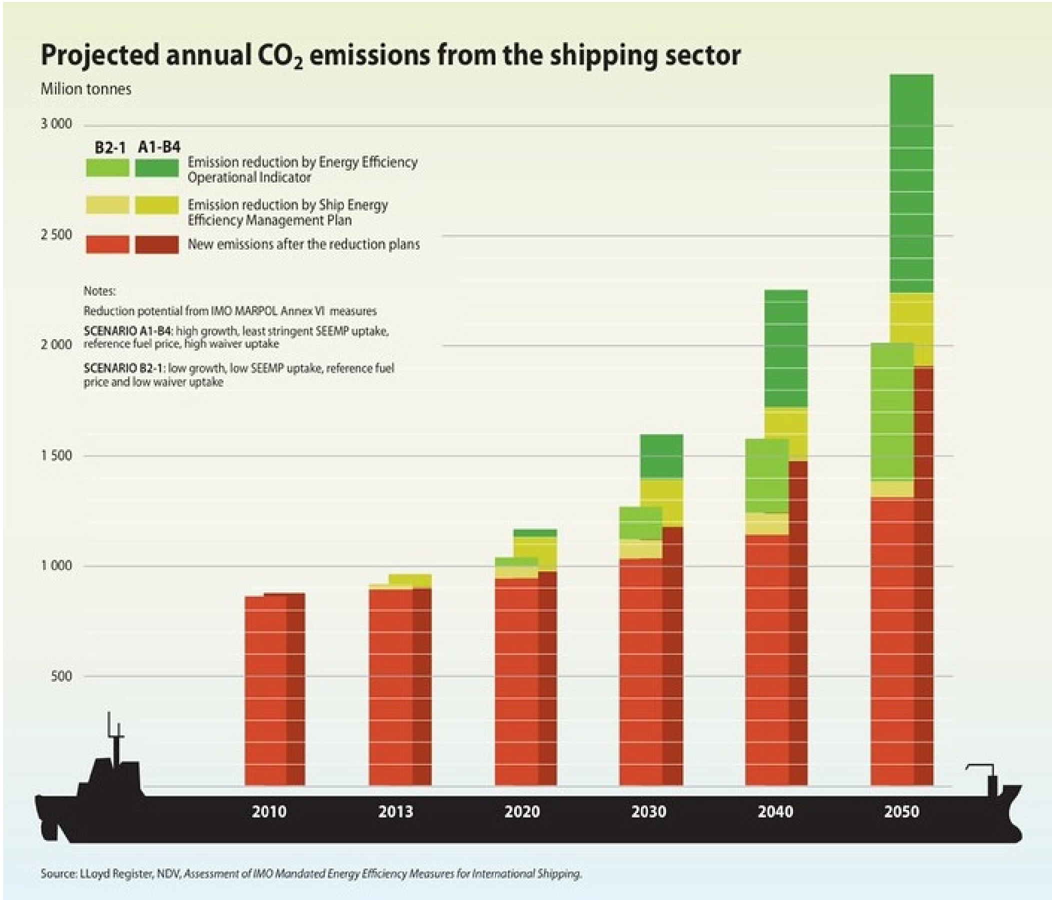


Figura 1.4: IMO greenhouse gas emissions strategy. Source: Lloyd Register, NOV, Assestment of IMO Mandated Energy Efficiency Measures for International Shipping

Hybridization of the propulsion chains are based on the separation or simultaneous use of several different energy sources. The choice of different sources, and sources combinations is designed to adapt to the ship power and energy requirements which include the needs of low and high power, the needs of flexibility in changes of rhythm and the needs of redundancy and reliability.

There are three main architectures for hybrid propulsion chains.

- Series Hybrid architecture
  In this type of HPS architecture, two kind of energy system (combustion engines and electric systems) are used in series. The propulsion is done by a variable speed electrical drive (propulsion motor and power converter). The electrical power of the propulsion motor is generally supplied by a set of generators driven by combustion engines. Another option is to use fuel cells to provide electricity or a combination of fuel cells and generators driven by combustion engine. Energy storage systems (ESS) (battery, supercapacitors, etc.) can be connected to the electrical bus to provide propulsion power. This ESS can be used separately or with the generators. In this configuration, electricity is used as an energy vector between the combustion engine and the propulsion motor, so there will be no direct mechanical connection between the engine and the propeller. This configuration allows dissociating the operating points of the combustion engine and of the propeller. This is why

it allows increasing the global efficiency of the system by optimizing the operating point of combustion engine and propeller in efficiency point of view. Of course, this advantage would be more significant if several generators and ESS are used, because it allows providing the required propulsion power by choosing an optimal combination of sources in order to keep the running combustion engines in their optimal operating area. However, for small ships, the possibility of multiplying the sources is very limited by volume and mass constraints.

- Parallel Hybrid architecture
  Parallel hybrid architecture combines two kinds of propulsion motors: electric motors and combustion engines which are mechanically coupled by the same shafts through gearboxes and clutches. The electrical motors and drive are supplied by electrical ESS and/or independent power sources generators driven by combustion engines and/or Fuel Cell for example. The two propulsion systems can be used together or separately. In most cases one of the propulsion systems is devoted to be used for low speed operations (classically the electric system) and the other one (classically the combustion engine) is used for high speeds. The first system can also be used as a boost system to provide a supplementary power during transient (starting, acceleration, etc.). This architecture reduces the number of elements compared to hybridization series and allows optimizing the sizing of each of the energy sources.

Among vessel types, tugboats—used for towing and maneuvering large ships—are among the highest emitters per unit of energy due to their highly variable load profiles [3]. Tugboats and ferries are ideal candidates for hybridization due to their operational profiles, which involve prolonged idling, low-speed maneuvering, and frequent speed changes that lead to inefficient fuel consumption [9].

This paper focuses on designing and sizing an energy storage system for a hybrid tugboat as a specific electro-mobility solution. Despite advancements in marine hybrid technologies, a standardized methodology for tugboat hybridization remains undefined [10]. The study presents a design methodology addressing parameters such as load profiles, power and energy demands, battery technology selection, and propulsion system optimization. Theoretical analysis is presented and simulation shows a emissions reduction while maintaining the robustness needed for towing and transit activities.

1.2. Challenges and Research Opportunities

1.3. Thesis Objectives and Outline

[1],

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